

EXP 10

NETWORK SLICING ENVIRONMENT USING MATLAB

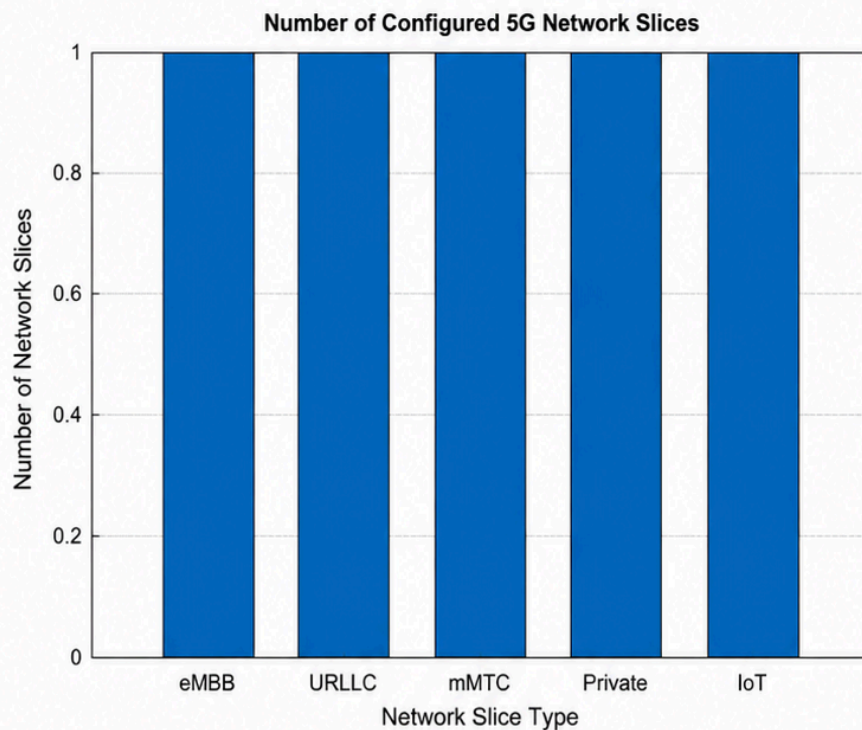
OBJECTIVE:

Analyze the Number of Network Slices.

CODE:

```
clc
clear
close all
slices = [ 1 1 1 1 1 ]; % One instance of each network slice
count = [ 1 1 1 1 1 ]; % Count of configured 5G network slices
figure
bar count
grid on
set(gca, 'XTickLabel', slices)
xlabel('Network Slice Type')
ylabel('Number of Network Slices')
title('Number of Configured 5G Network Slices')
```

OUTPUT:

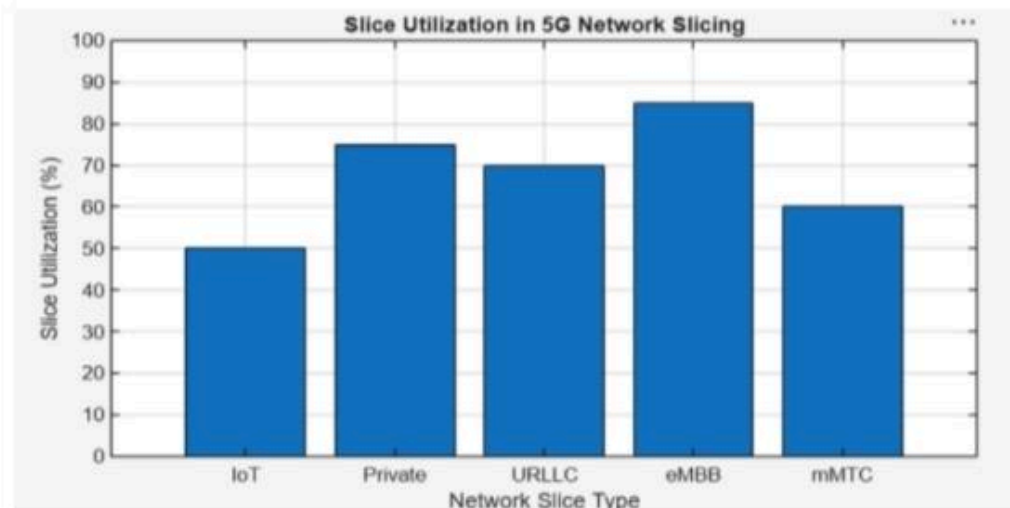


Objective: Compare Slice Utilization (%)

CODE :

```
1 clc;
2 clear;
3 close all;
4 slices = categorical({'eMBB', 'URLLC', 'mMTC', 'Private', 'IoT'});
5 utilization = [85 70 60 75 50];
6 figure;
7 bar(slices, utilization);
8 grid on;
9 xlabel('Network Slice Type');
10 ylabel('Slice Utilization (%)');
11 title('Slice Utilization in 5G Network Slicing');
12 ylim([0 100]);
```

OUTPUT :



Objective Analysis of 5G Network Slicing End-to-End Latency (ms)

C O D E :

```
1  clc
2  clear;
3  close all;
4  slices = [1 2 3 4 5];
5  latency = [25 5 40 15 60];
6  figure;
7  plot(slices, latency, 'bo', 'LineWidth', 2, 'MarkerSize', 8);
8  grid on;
9  xticks(slices);
10 xticklabels({'eMBB', 'URLLC', 'mMTC', 'Private', 'IoT'});
11 xlabel('Network Slice Type');
12 ylabel('End-to-End Latency (ms)');
13 title('5G Network Slicing: End-to-End Latency');
14 ylim([0 70]);
```

O U T P U T :

